# 16-220 Robot Building Practices - Final Design Report

16-220 Robot Building Practices | **Andrew ID:** derekpen | **Name:** Derek Peng

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## Report Purpose

This report documents your complete robot design, engineering calculations, development process, and learning outcomes. It is organized in three main sections:

1. **Final Robot Design** - What you built (technical description, CAD, analysis)
2. **Design Evolution, Milestones & Demonstrations** - How you got there (iterative testing, failures, improvements)
3. **Critical Analysis & Reflection -** What you learned (proposal vs. reality, lessons learned

## Submission Requirements

- **CAD requirement:** Must include multiple CAD screenshots showing design evolution and final CAD of your system as well a link to your CAD. CAD should ALSO be submitted through the 16-220 onshape classroom assignment.
  **CAD files must be named as** “First Name_Last Initial_16220 2025 Final Robot” ie: “Zeynep_T_16220 2025 Final Robot”
  All items in CAD should be correctly mated, part studios and assemblies should have defined names that are easy for a reviewer to understand.
- **Documentation:** Must include photographs/visual evidence of the final robot and evidence of thought throughout iterations.
- **Professional quality:** Clear figures with captions, proper formatting, proofread
- **Document shall be submitted as a pdf to Gradescope No later than December 12, 2025 5:00pm**

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# Section 1: Final Robot Design (38%)

## 1.1 Executive Summary (5%)

*Briefly summarize your project (1-2 paragraphs) including:*

- *Your final robot strategy and approach and key differences to approach from your initially proposed solution.*
- *Key mechanisms and innovations of your robotics system.*
- *Final competition performance and score.*

My final strategy was to use a ramp to climb onto the board, then use a passive, flexible claw to manipulate the bean bag and push it into the hole. This differed largely from my original proposal, as I decided to use my backup plan of using a ramp to climb the board instead of my “whegs”. Furthermore, my method of acquisition was vastly different as I used an unpowered claw, compared to my proposed designs of a roller system or powered claw.

My key innovations and mechanisms in my system are the ramp and passive claw. The ramp provided a smooth and consistent way for my robot to get on and off the ramp. Meanwhile, the passive claw removed any need for extra actuators, resulting in a much more simple and elegant design while still maintaining a high degree of reliability.

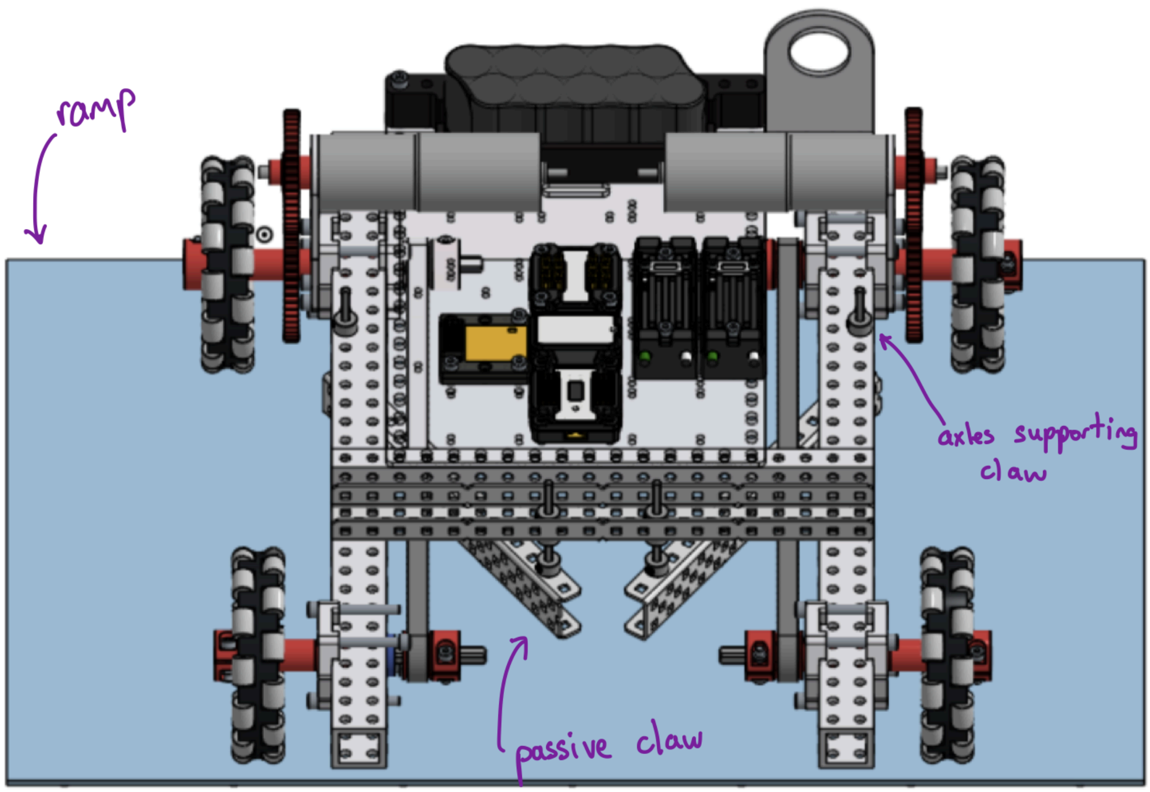
My final competition performance overall was 15 bean bags scored in 4 minutes and 29 seconds.

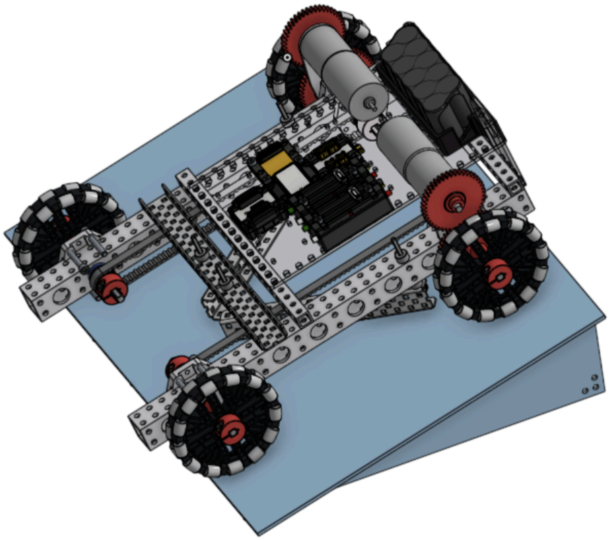
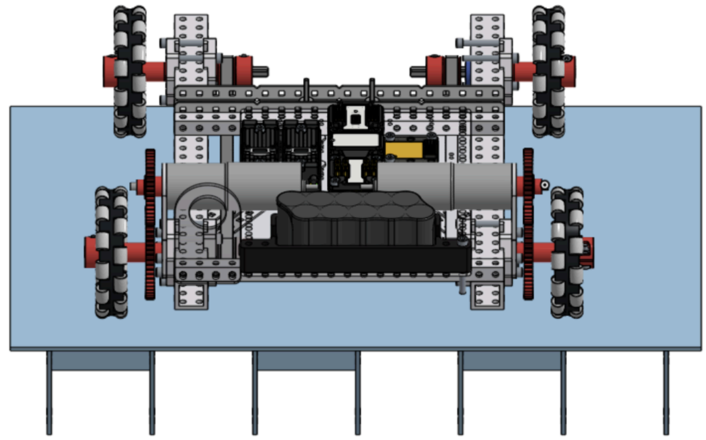
## 1.2 System Overview (10%)

**Physical Description:**

- *What is overall configuration or your robot? Explain with labeled CAD renderings from multiple views.*
- *What are the key dimensions of your robot at the starting configuration (show you meet 2'×2'×2' starting constraint) and during operation?*
- *How does your robot accomplish the task? Provide a high-level description.*

The overall configuration of my robot is split into two parts: my ramp and the robot itself. My ramp is stationary, and stays at the base of the game ramp, while the robot is free to move around to score points.



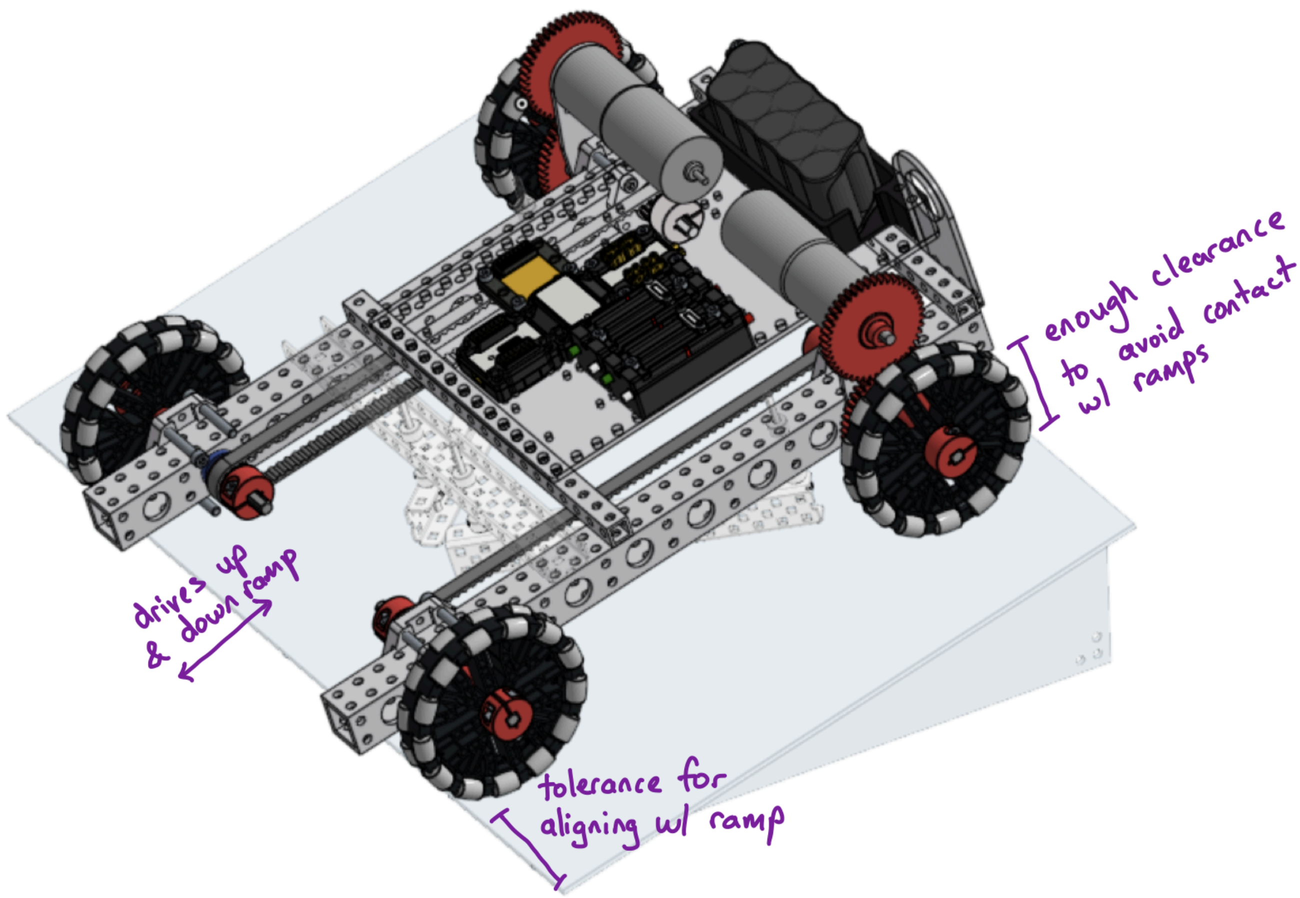
The dimensions of my robot at the start are 15”x21”x10”. During operation however, my robot “expands” since my ramp is left behind as my robot moves - where my ramp has dimensions of 14”x21”x4” and my robot has dimensions of 16”x14”x6", which does not change when manipulating a bean bag.

My robot accomplishes the task by simply driving into a bean bag, which allows the passive claw to latch onto the bean bag. Then, the robot drives up my ramp, pushing the bean bag onto the game ramp, where it can then drop it into the hole in the game ramp. Finally, I can drive off the ramps in reverse and continue the cycle.

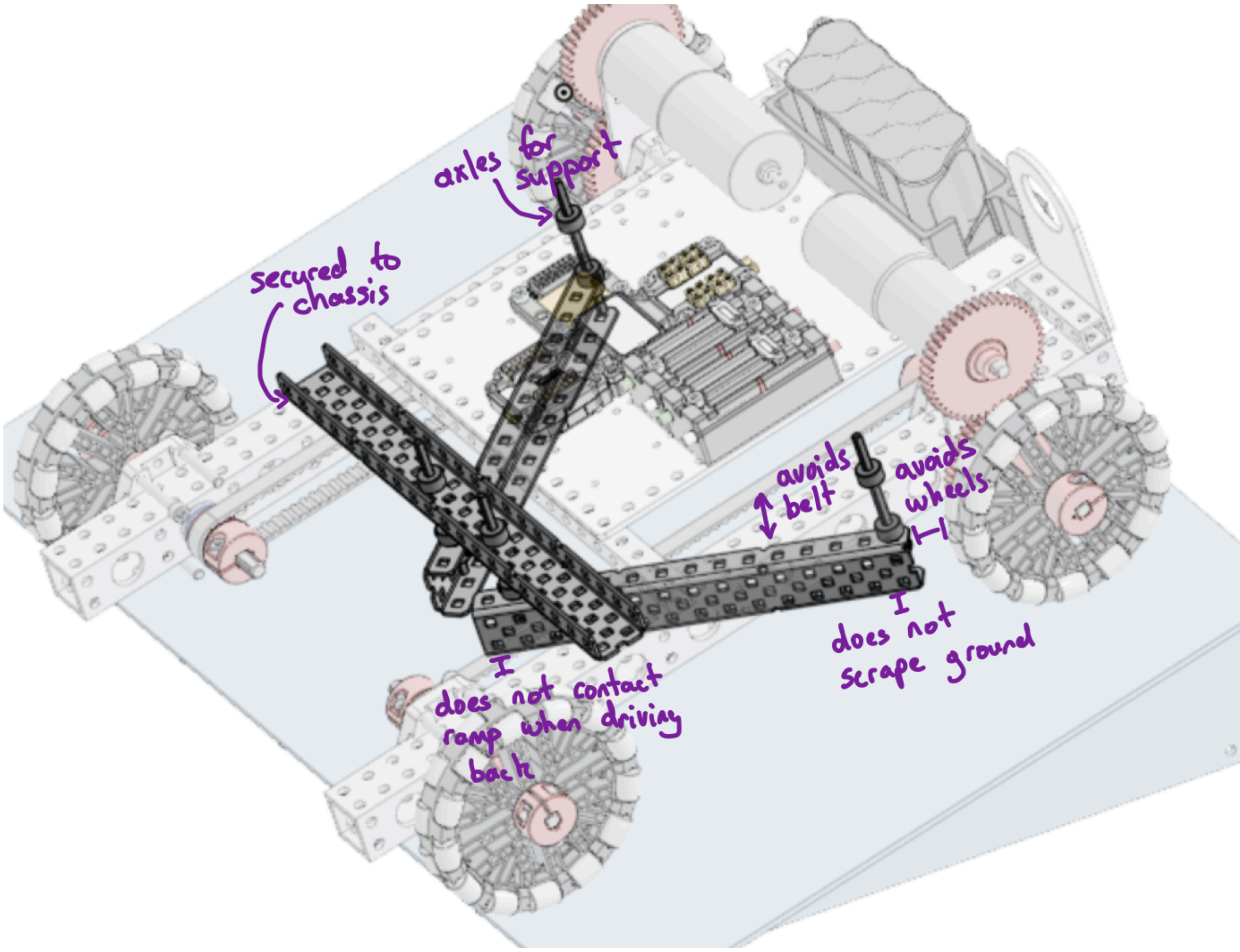
**System Architecture:**

- *Provide a pictorial diagram showing major subsystems and their interactions.*
- *List major subsystems: chassis/mobility, intake, storage/indexing, aiming, launch/scoring*

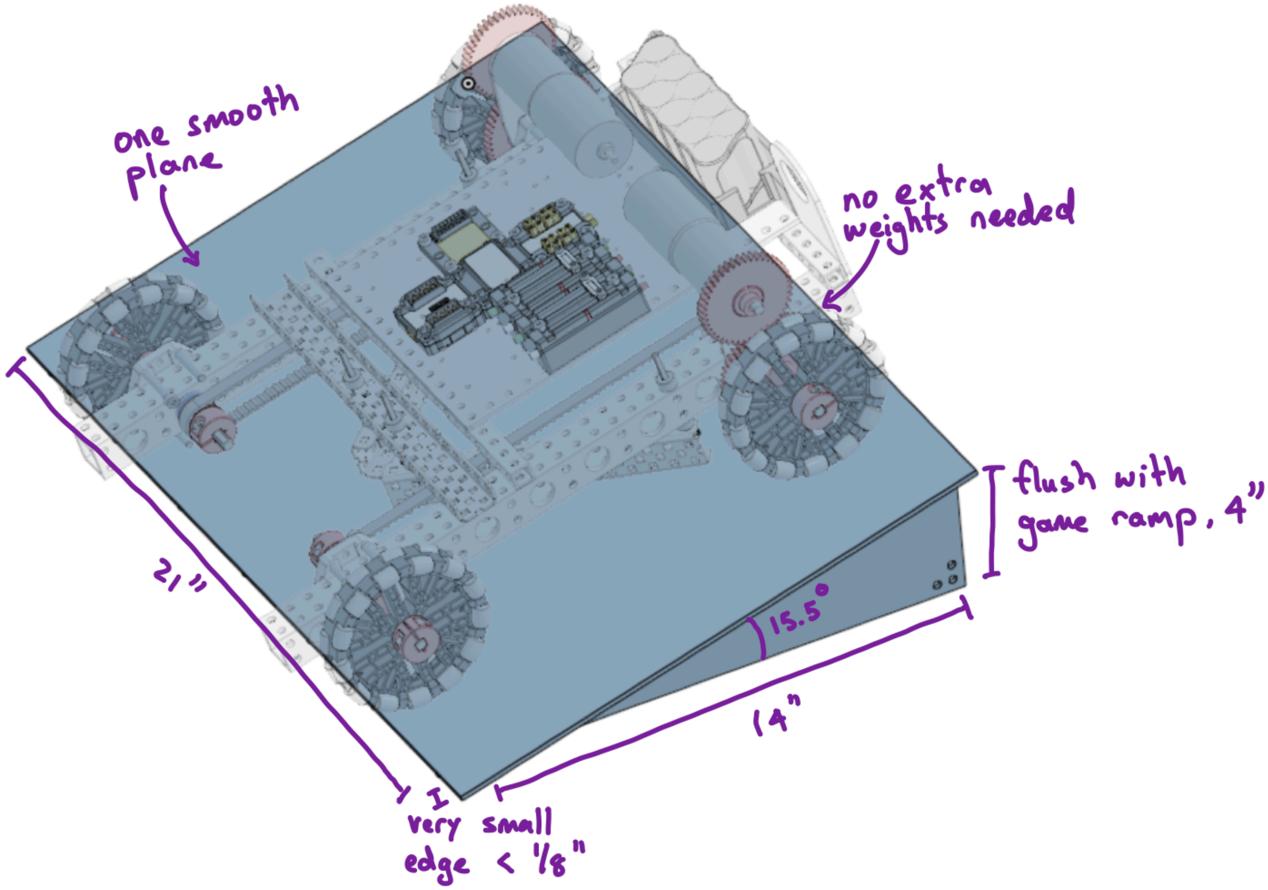
Chassis:



Intake:


axles for support
secured to chassis
avoids belt
avoids wheels
does not scrape ground
does not contact ramp when driving back

Ramp:




***Required:*** *Full robot CAD assembly with annotations identifying key components (insert OnShape link, and also submit OnShape through the "16-220 OnShape classroom assignment")*

**https://cad.onshape.com/documents/1b1a7505c4f1f56befa2eda1/w/bfbc162863a70b543faaf860/e/977b742970d0d73eddfa8fad?renderMode=0&uiState=69389551057739d39e68c863**

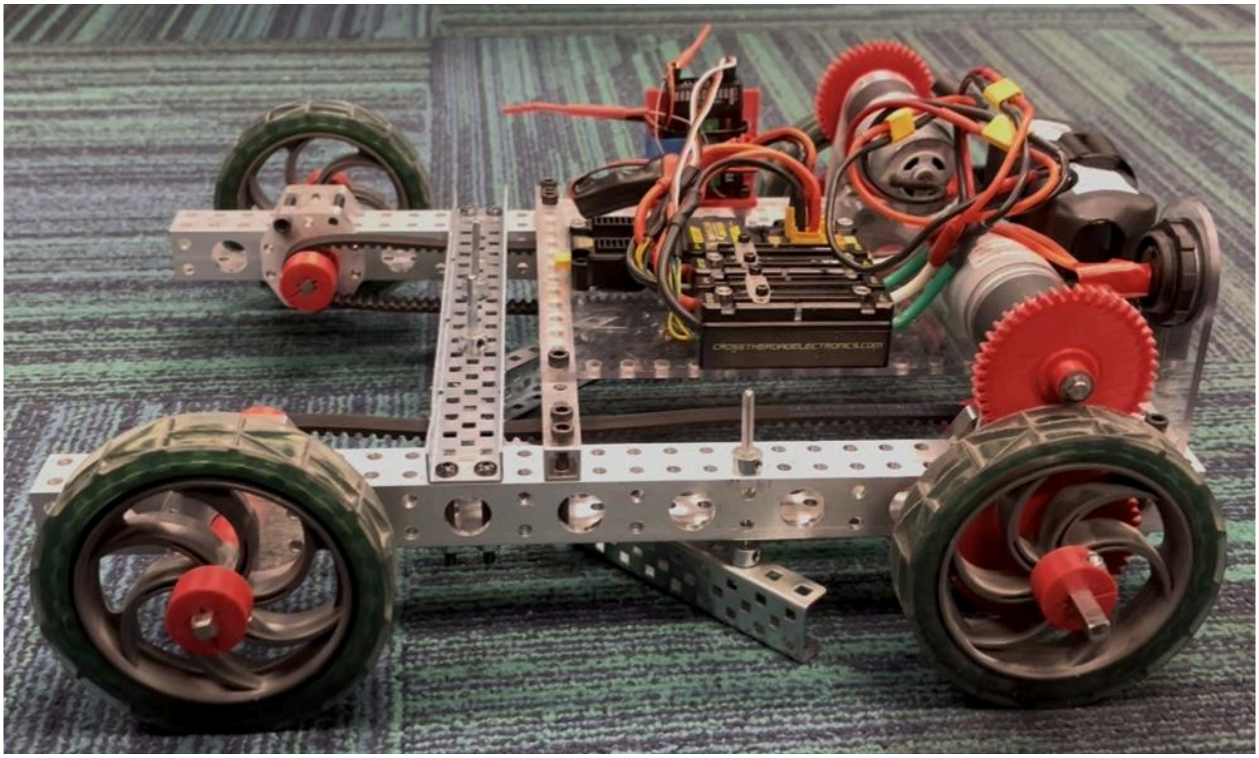
## 1.3 Subsystem Detailed Designs (23%)

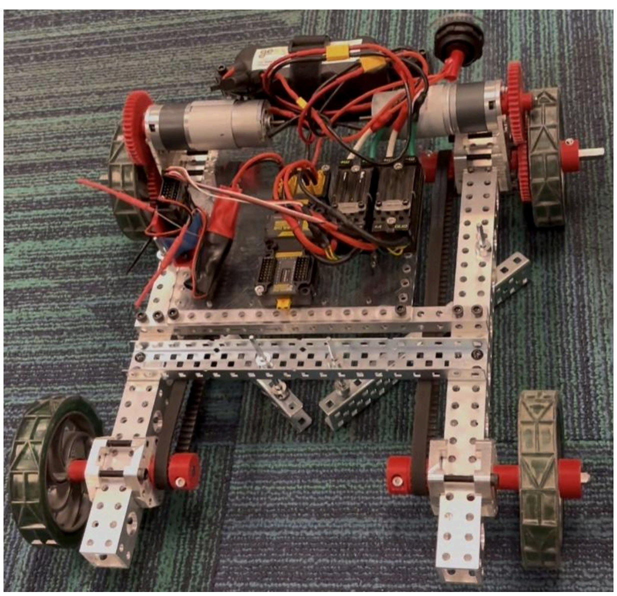
**For each major subsystem (intake, scoring mechanism, chassis/drivetrain etc.), provide the following:**

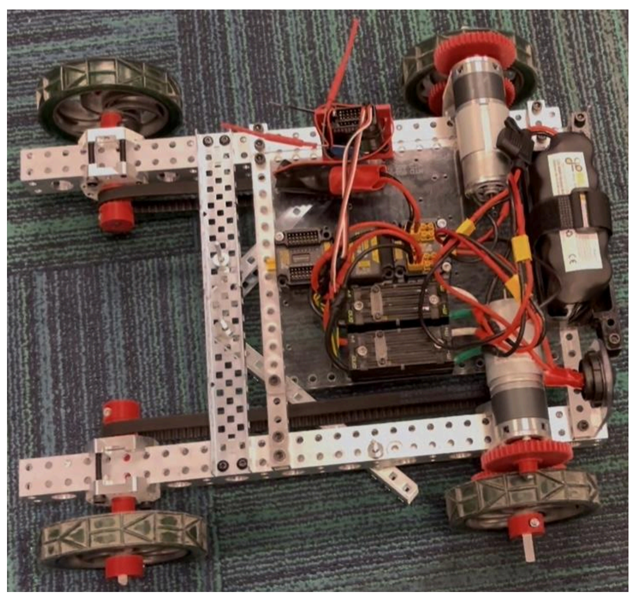
**Mechanical Design:**

- *CAD drawings/screenshots with dimensions and key parts labeled*
- *Images of physical as built systems*
- *Description of how system works (mechanism operation)*
- *Key components: motors, servos, gears, linkages, structural materials*
- *Manufacturing method rationale and materials rationale for each component: why you selected "this" manufacturing method compared to the other methods you had at your disposal (ease of manufacturing outsourcing to a printer/router queue or selection from vex stock should not be the single primary rationale).*

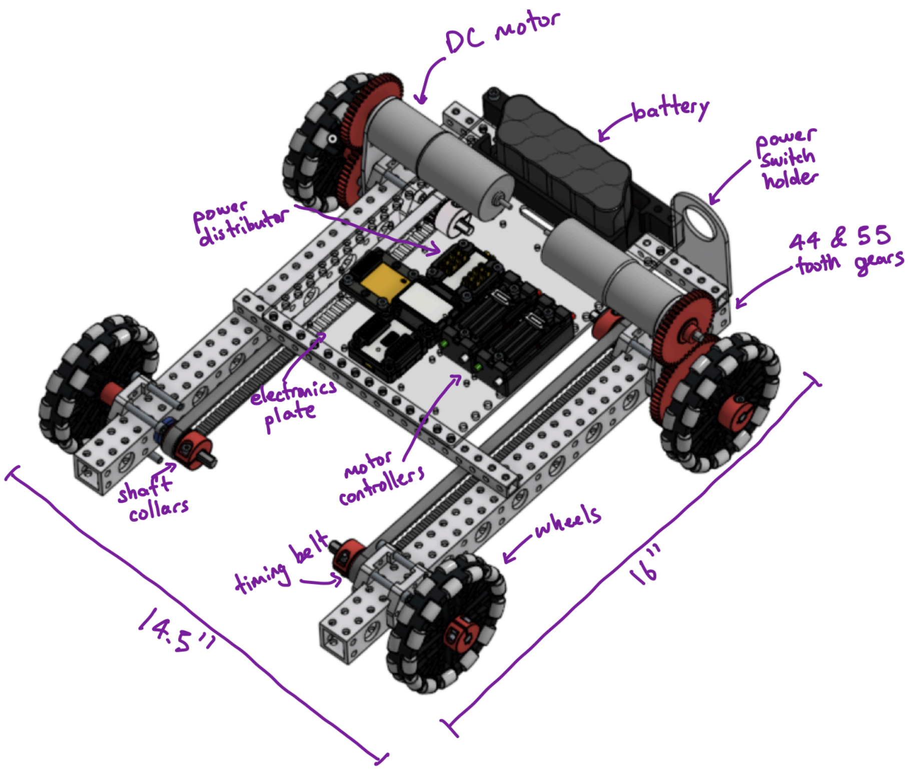
Physical System Photos:





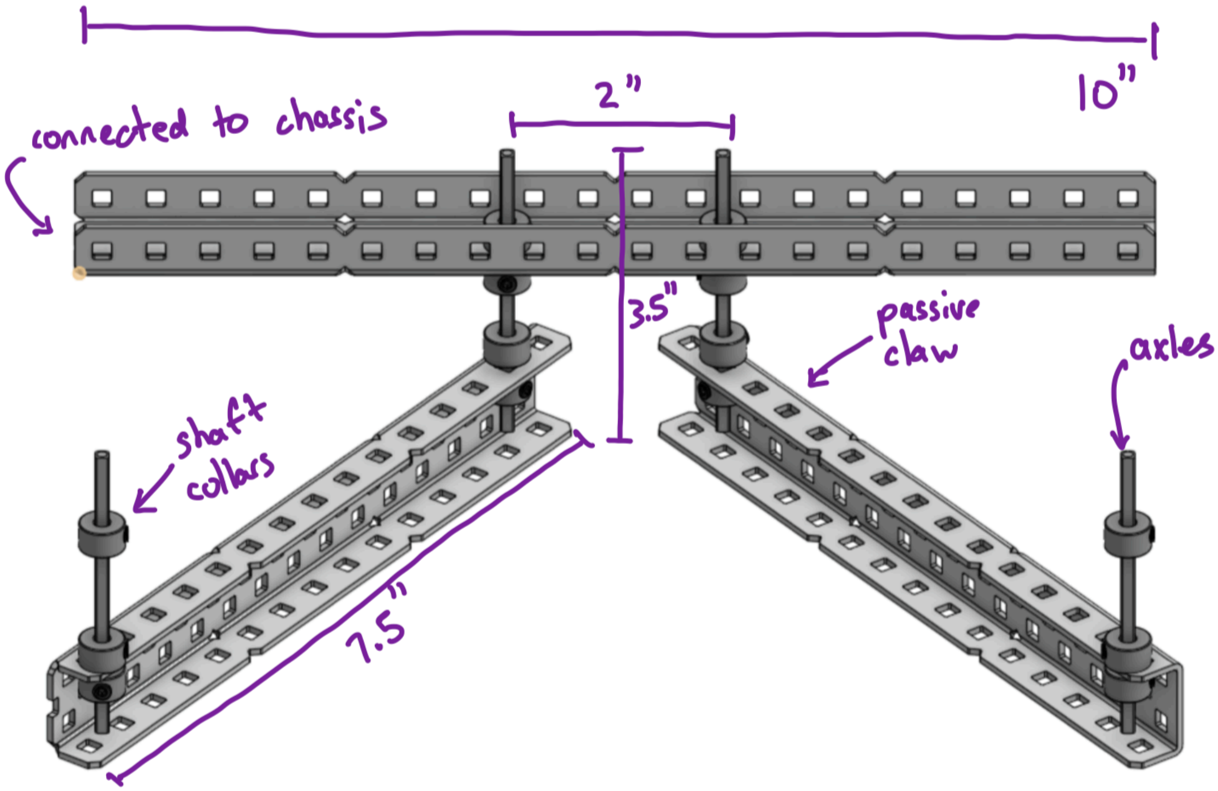


Chassis:




The chassis is powered by a 12V NiMH battery, distributed by the XT-30 board. For movement, I used the NFP-GA36Y-555 motors on either side of the chassis to individually move each side of the robot. The motors have a 44:55 gear ratio, and the back wheels are connected by a timing belt.  The manufacturing of this robot is through the instructions provided.

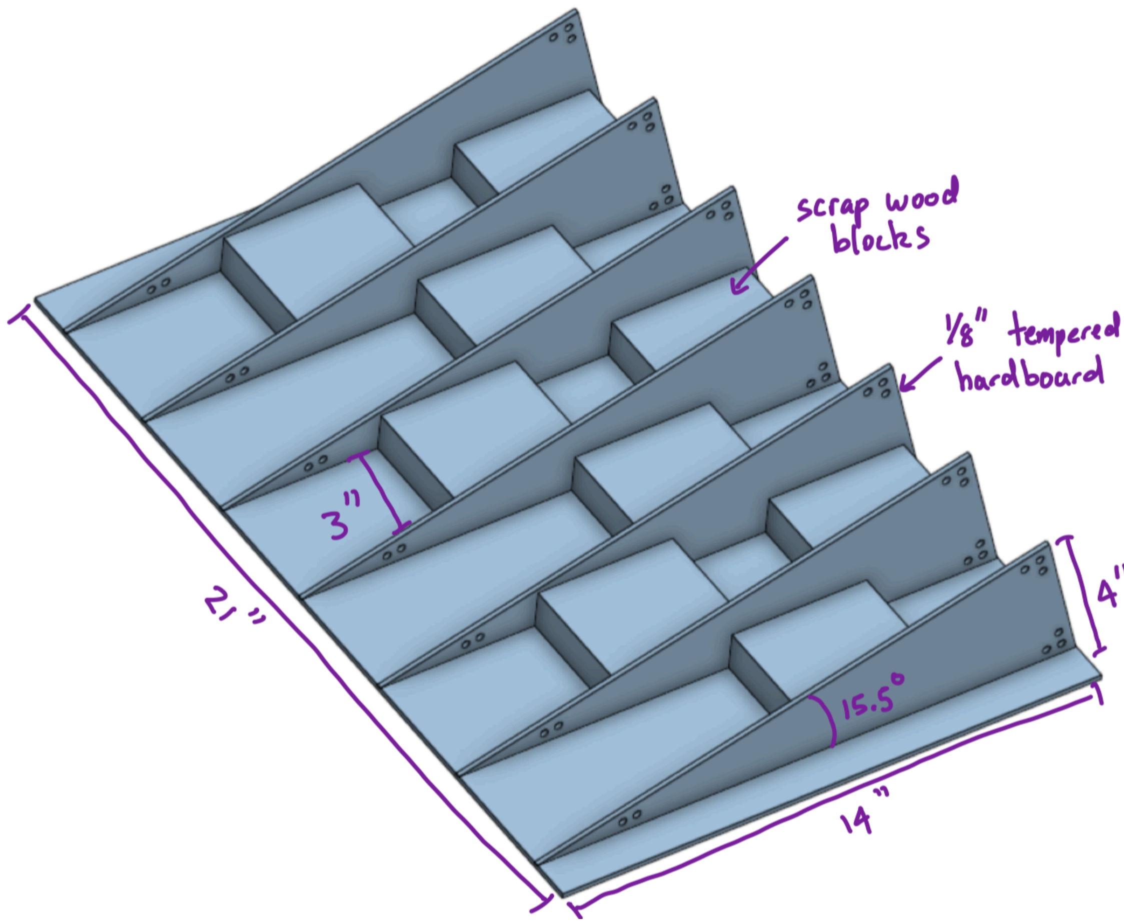
Intake:




The passive claw works by being able to freely move and adapt to the shape of the bean bag, allowing it to have a consistent hook on it. The axles replace traditional screws, as the axles are smaller than the holes in the c-channels, which gives it room to move around. The shaft collars are strategically placed to either prevent the shaft from falling through or to restrict vertical movement. This means no actuators are needed.

Every part used here is Vex stock, so there was no manufacturing process involved.

Ramp:



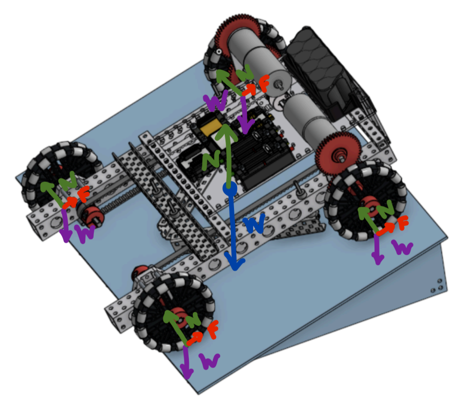

The ramp is made out of tempered hardboard and scrap wood blocks, held together by glue and nails. The ramp works by sitting flush against the game board, providing a path for my robot to drive over.

To manufacture it, I used a vertical bandsaw to cut out each triangle. For the large panel, Tim helped me use a table saw to cut it out. For the individual wood pieces, Tim showed me how to use a miter saw to cut each one to 3”. Afterwards, I used a combination of glue and nails to secure each triangle to a block, and similarly to secure the large panel to the triangles. The reasoning behind the process stemmed from the fact that the cuts I needed were relatively straight and simple cuts. I did not need any circular or curved cuts, making it very easy to simply use saws to cut everything. As for putting it all together, that was mostly Tim guiding me through how to use new tools like the nail gun.

**Design Calculations & Analysis:**

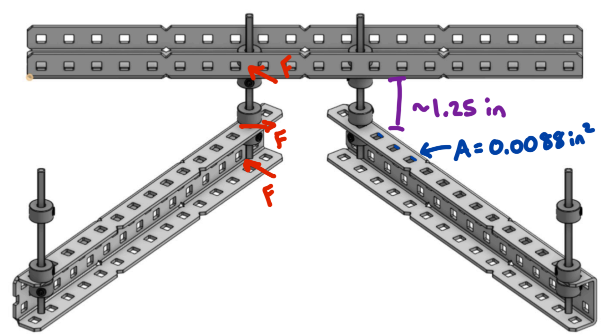
- *Free body diagrams showing forces and loads (These should be done for full system and subsystem levels, evaluating the loads on each shaft, gear, linkage arm, motor/servo)*
- *Motor/servo selection justification with torque/speed calculations*
- *Gear ratios and mechanical advantage analysis*
- *For projectile systems: trajectory analysis, launch velocity calculations*
- *For ramps: angle+material selection, and design considerations.*
- *For linkages: kinematic analysis, range of motion*
- *Material selection justification*
- *Calculation of forces through gear train and or linkages, culminating on the final force on either the square shaft or D shaft of the driving motor (i.e., what is the load on the area of the D shaft or square shaft motor interface, can the materials in the interfaces withstand these interaction forces without deforming? What is the factor of safety?).*
- *For each cantilevered load in the subsystem calculate the energy loss and deflection of the shaft/structure based on its material properties. Explain how system performance and long term reliability is affected by this. Include diagrams.*
- *Justify the selection of the axle shaft size and type ( ⅛ square, ¼ square, ⅜ hex etc)*

Whole System Analysis:




For the whole system at the starting configuration, the weight of the robot, excluding the ramp, is opposed by the normal force and friction at each wheel. Then, the weight of the entire system is spread out through the seven base triangles and pushed onto the floor, which is again opposed by the normal force.

Intake System Analysis:




The main force present in the claw is when a bean bag is being pushed. This causes the bottom half of the c-channel to deflect backwards, resulting in the top half pushing into the axle above. This is then translated to the top of the axle where it is connected to the top c-channel. Therefore, the maximum force felt is at the top, which is about 1.25 in from where the force of the bean bag is applied. The area of contact between the shaft and c-channel is about 0.0088 in$^2$. Using the 1018 steel data sheet, the yield strength is 53700 psi, which can be used to calculate an upper bound of the force that can be applied: $53700 = \frac{F*1.25}{0.0088}$, $F = 378\, lb = 1681\, N$. This far exceeds the force of anything in the system, and so the axles supporting the claw should never break, even after repeated use.

Ramp System Analysis:

As shown above, the ramp has a width of 21”, length of 14” and a height of 4”. It was these constraints that led to the ramp having an angle of 15.5 degrees.

The material I chose was ⅛” tempered hardboard to build the ramp since it could be laser cut and assembled together quickly. The main motivation for this was my experience with this material in the class Fundamentals of Mechanical Engineering (24-101), where we laser cut this exact material to build a hand powered flash light. This material was cheap, lightweight, rigid, and had both a smooth and rough side, which I had planned on using depending on whether or not I need more or less grip.

As for the design considerations, I had to keep two things in mind: fitting the ramp inside the storage bins and fitting the robot and ramp into the starting constraints. The storage bin had the biggest constraints; the bottom of the bin measured 20.6” x 16.6”. While the bin did have a larger area at the top, I still had to make sure the ramp fit with the robot and controller box. Therefore, the width of the ramp was 21” to be able to snugly fit inside the bin at an angle. Meanwhile, the length of my ramp was only 14”, a few inches shorter than the bin itself. This was motivated by the fact that I was not too sure if the robot could actually stay fully on top of the ramp without sliding down during the starting configuration. Therefore, the length of my ramp was shortened to accommodate that need if it arose. As for the height of the ramp, I measured the bottom edge of the ramp to be 4” high, so I wanted the ramp to be flush with the game ramp. These constraints resulted in the ramp having an angle of 15.5 degrees.

<u>Chassis Analysis:</u>

For the weight of a chassis, I’ll choose it to be at 20 pounds, which is a very high estimate of its actual weight. Therefore, assuming the weight is distributed evenly, each wheel experiences 5 pounds of force. Again, as an upper bound, I assume that the motor will need to overcome the 5 pounds of force, and not the friction needed to move the robot. This means all of the forces and torque calculated will be much higher than the actual forces and torque experienced by the robot. The wheels are 4 inches in diameter, resulting in a torque of $5 * 0.1667 = 0.833\, ft.lb = 10\, in.lb$. The wheel contacts the axle with an area of approximately 0.128 $in^2$, resulting in a pressure of $\frac{0.8333*12}{0.128} = 78.1\, psi$. Assuming the axle is made of 1018 steel, where its yield strength is 52700 psi, this means the drive axle should never break, with a factor of safety of $\frac{52700}{78.1} = 675$.

Then, the shaft is connected to the 55 tooth gear, with a diameter of 2.75 inches. The 55 tooth gear meshes with the 44 tooth gear, exerting a torque of $10 * \frac{2.75}{2} = 13.75\, in.lb$. The area of the teeth face is around 0.0264 $in^2$, resulting in a pressure of $\frac{13.75}{0.0264} = 521\, psi$. Looking a the data sheet for PETG, the tensile strength of it is $34 \pm 4$ MPa. Since we want the maximum shear strength, we can multiply it by 0.6 to get an approximation for shear strength (found on Google), resulting in $30\, MPa * 60\% = 18\, MPa = 2610\, psi$, using the lower bounds of strength. This results in a factor of safety of $\frac{2610}{521} = 5.00$.

Lastly, the 44 tooth gear transfers its torque to the motor shaft. The 44 tooth gear has a diameter of 2.2 inches, so the torque needed is $13.75 \times 1.1 = 15.125\ in.lb$. Looking at the drivetrain motor data sheet, it has a maximum torque of $100\ kg.cm = 86.8\ in.lb$, meaning the torque needed to drive the entire drive train is much lower than the motors maximum torque.

**Control Strategy:**

- *Explain the usage of RC channel(s) to control this subsystem*
- *Justify channel selection. (Why did you set up your controller this way? How did it benefit you? How did it hinder you?)*

I used channels 1 and 2 to control the left and right part of the drivetrain with the right joy stick, which made driving more intuitive for me. I set the channel powers to 80% to make the drivetrain less sensitive. This allowed me to have a smoother control over the robot, especially when driving up and down the ramps. I did not need to modify the gear ratio to get my desired speed reduction as the extra torque was not needed.

The design of my scoring mechanism meant I did not need any other actuators, and so had no need for the other channels. This greatly benefitted me as it meant I could focus all my attention on driving and did not need to move my hands around to control other motors or servos.

## Section 2: Design Evolution, Milestones & Demonstrations (32%)

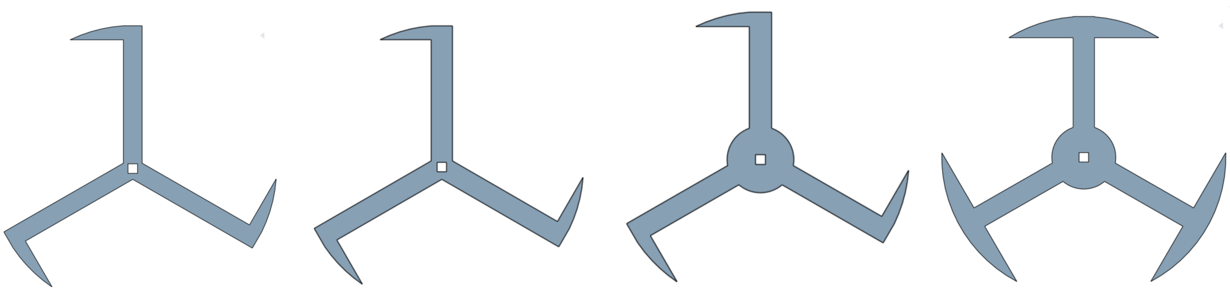
**This section documents your iterative design process through each demo milestone. For each demo, show how testing and analysis drove design improvements.**

### 2.1 Demonstration 1: ~~Intake System~~ Getting onto Game Ramp (8%)

**Initial Design Concept:**

- *Provide early sketches or CAD of intake (version 1.0)*
- *Explain design intent and expected performance*
- *Explain why this approach was chosen*

My initial milestone was altered from demonstrating an intake system to getting onto the game ramp. The reasoning behind this was mainly due to the fact that my initial design with the “whegs” heavily impacted how other subsystems would work. The whegs completely changed the robot’s gait, making its vertical height oscillate as it moved. This made it unfeasible to use rollers, or other forms of stationary intakes, to manipulate the bean bag. As a result, it was more important to test the whegs before trying to build an intake.



*Version 1* *Version 2* *Version 3* *Version 4*

My initial idea with the whegs was to use these spoked wheels to hook over the edge of the ramp, which would lift the robot up and onto the ramp. This was an interesting traversal mechanism I found on Youtube, where they used four spoked whegs, as highlighted in my design strategy proposal: [https://www.youtube.com/watch?v=stiSSy37LWc](https://www.youtube.com/watch?v=stiSSy37LWc)

The design I made involved only three spokes, which meant I could have a smaller wheel diameter and still have the same climbing effectiveness as a four spoked wheg. I measured the vertical height of the game board at the bottom to be around 10 cm, and so decided to make my wheg diameter also 10 cm. Theoretically, I did not need the diameter of the wheels to be 50% of the climb height. I could have decreased the diameter, but because of the incline of the game ramp and the edge, I decided to make it bigger to be safe. The downsides to this design choice were that lack of torque would be a bigger issue, and the problem with the gait and vibrations would be further amplified compared to a smaller wheg. To manufacture it,

I 3D printed it with PLA because it would be light, accurate and easily replicated. Other factors included the need for it to be relatively thick, and the complexity of the shape required precision as it needed to be a perfect circle on the outside edges.

This design was chosen because it seemed like a niche yet elegant way of solving the problem of getting onto the ramp. In hindsight, I should have realised that torque and gait would be huge problems, especially their effects on other subsystems.

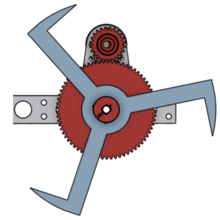
**Testing & Results:**

- *How did you test your system? Explain your setup and methodology.*
- *What worked well?*
- *Did you identify any failure modes and/or parts/components that underperformed? How?*
- *Provide video/photo evidence for your test and system.*
- *What was your plan to make your system perform better in the next demonstrations?*

To test the whegs, I first printed two wheels and attached them to the front. Using normal omni-wheels at the back, I tried to climb the ramp. It was able to lift the front part of the robot over the game ramp, but got stuck as the smaller omni-wheels made the drivebase contact the edge of the game ramp. This seemed promising, so I decided to print two more to lift the back half up.

When testing with all four wheels replaced with whegs, the robot was able to climb the ramp completely, However, it was very prone to slipping, so it could take several seconds of grinding against the game ramp to actually get on. It was also very difficult to move around on the ramp, especially when I tried to get off the game ramp. Another issue was that I could not go backwards consistently. The robot's weight distribution was very forward heavy to make it easier to get on the ramp, but this caused the robot to flip over if I tried to go backwards.

To solve these issues, I needed to increase grip and torque. To increase grip, I tried adding heat shrink to the tips as these were the critical points where the whegs contacted the game ramp. This only slightly increased the consistency of the robot when trying to climb the ramp. Then, I changed the gear ratio from 44:55 teeth to 24:70, resulting in about 2.5 times more torque. This change did significantly improve my mobility as it allowed me to drive backwards consistently, although trying to do so quickly would cause the robot to flip.



Another issue I encountered was that the intersection of the spokes was relatively weak, and so one of the whegs actually snapped when testing. To reinforce it, I extruded a circle around the intersection, making it much less susceptible to breaking (see Version 3 above).

After the demo, I tested two more things: using TPU instead of PLA and making a more circular wheg. Since the heat shrink did not make getting onto the ramp consistent enough, I decided to print two more whegs made out of TPU to test out. After numerous trials with them on the front, back, and alternating them with the PLA whegs, it performed the same or even worse as with the heat shrink grippers. Even combining both of these methods did not yield good results.

Lastly, because the gait was so unstable and made it incredibly hard to maneuver, even on flat ground, I tried to make a more circular wheg that was closer to a normal wheel (see Version 4 above). It was the same design as Version 3, except I mirrored the protruding arm to have it on both sides. To ensure I could still climb the ramp, I made the protruding arms a little thinner. With this new design, the gait of the robot wasn't changed much, mainly because I had only printed two to see how it would work in conjunction with the Version 3 wheg. Seeing as though the robot still could not consistently climb the ramp, I chose not to print more copies of the new wheg, and chose to pivot my design strategy, as outlined below.

**Key Learning Takeaways:**

- *What did you learn about the materials you were making your robot from?*
- *What did you learn about manipulation of the bean bag? (explained later)*
- *According to your learnings, what was your evaluation of your intended system?*
- *In light of your learnings, did you change your approach in designing and building your robot?*

From my testing, I learned that PLA was a very rigid material that was susceptible to fracturing under repeated and sudden forces. I had to modify the design of my whegs in an attempt to distribute the force more evenly, like the addition of the centre circle in Version 3. Furthermore, I learned about the properties of TPU and how it was flexible and more grippy than PLA. Unfortunately, the material did not provide enough traction for my intended design.

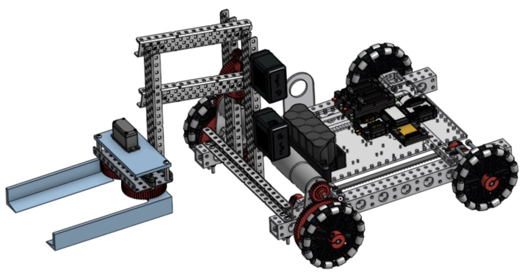
After going through several iterations of the wheg and related components (like drivebase gear ratio), I decided that my whegs were just not good enough. It lacked consistency, mainly due to grip issues, and it introduced an inconsistent gait that made it harder to design other subsystems. Additionally, the shape of the whegs made turning with precision almost impossible, especially if I was on the game ramp. Therefore, the implementation of the wheg was flawed and would not be suitable for this task.

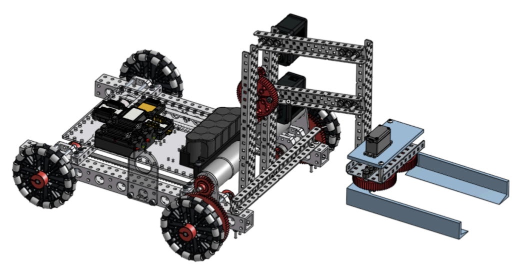
With all of this in mind, I decided to change my approach. As outlined in my design strategy proposal, I switched to my backup plan of using a ramp to get onto the game board.

## 2.2 Demonstration 2: Scoring System (8%)

**Initial Design Concept:**

- *Provide early CAD of scoring system*
- *Explain your design rationale including expected throwing distance or placement method*
- *Provide relevant calculations (motor speed, torque etc.)*





My initial intake design consisted of a servo-powered claw and a motor controlled fourbar to lift the claw. The expected performance was for the claw to grab the bean bag, then the fourbar could lift the claw, giving me space to climb the game ramp and then score the bean bag. This design was chosen because of my initial robot design, where I used “whegs” to climb the ramp. The alternative would have been to use rollers, but this would not have worked with the whegs, as the distance between the ground and bottom of the rollers would always be changing significantly due to the gait of the whegs. I had to use a system that enabled me to vertically move the intake system, while still keeping it horizontal to the ground. Therefore, I used a fourbar and claw to achieve this.

My main concern was whether or not the system had enough torque. For the servo powered claw, I chose the 2000 series servo with a stall torque of 350 oz-in (~2.5 Nm) because it had the largest torque out of the available servos. Since the servo does not directly lift the bean bag up (it’s only used to close the claws around the bean bag), there was no way I could get the exact applied torque. For an upper bound, I decided to assume the weight of the bean bag was the weight applied against the servo (16 oz), which gave me a limit of 0.56 m for the perpendicular distance of the applied force. The length of my claw was less than 20 cm long, well below my limit, but I still had to be careful since the torque could easily exceed my calculations if I tried to grip the bean bag tighter.

As for the motors for the fourbar, I regrettably did not do any calculations. This was one of the biggest mistakes I made, and definitely should have done some preliminary calculations before trying to build the fourbar. Instead, I iteratively tested out different gear ratios until I had a satisfactory one, as outlined below in my testing and results.

**Testing & Results:**

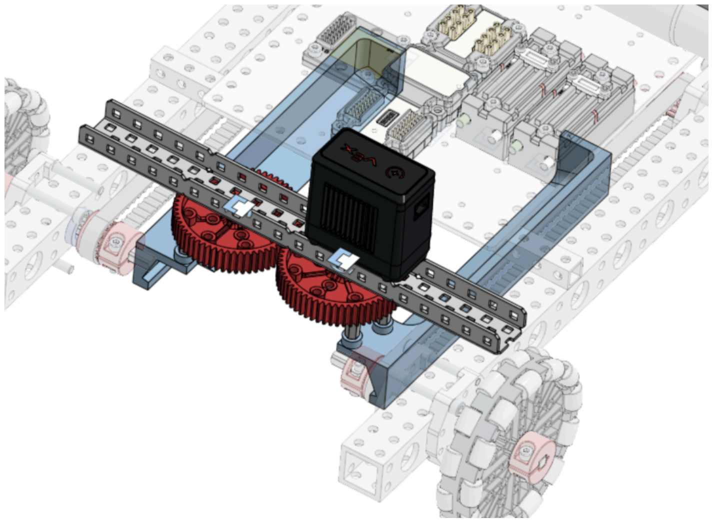
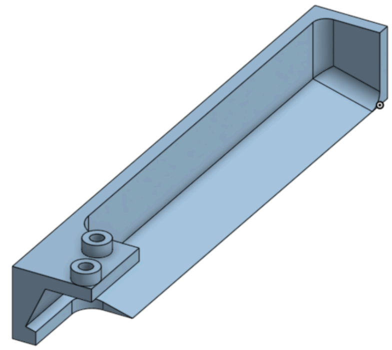
- *How did you test your system? Explain your setup and methodology.*
- *What worked well?*
- *Did you identify any failure modes and/or parts/components that underperformed? How?*
- *Provide video/photo evidence for your test and system.*
- *What was your plan to make your system perform better in the next demonstrations?*
- *Explain the problems and potential improvements in detail, e.g. insufficient power, inconsistency, gear failure, shaft interface failure, jamming, etc.*

I had two main testing phases. The first phase was to ensure the fourbar had enough torque to lift the bean bag and claw system. I iteratively tested gear ratios to see which ones were sufficient by ensuring that the fourbar could lift up and down at least three cycles. Since the bean bag and claw were heavy, I had to use the largest direct gear ratio Vex offered and two Vex motors. The second phase was then making sure the claw and fourbar could hold the bean bag while the robot moved around. To test this, I simply drove around with a bean bag in the claw, making sure to test turning and stopping.

Video Demonstration: Section 2.2 Demo.mp4

The claw ended up working well, having a tight hold on the bean bag. However, the tolerance in the gears and servo meant that it was hard to precisely control the servo to avoid stalling it for too long. This meant I needed to clamp the bean bag then slowly loosen the servo's grip on it, which wasted a lot of time. Another small structural issue with the claw was that to get the claw flat on the ground, I needed precise spacings between the claw and the gear. Since the length of the screws available was limited, the claw was not fastened very securely to the gears, so continued use could easily loosen the mechanism. Meanwhile, the fourbar was quite unstable because it only had one plane of support at the bottom with a single c-channel. This caused the fourbar to bend slightly forward when holding the bean bag because of the extra weight. As a result, the bean bag had the potential to slip out from the front.

To fix these problems, I redesigned the claw to have a lip at the end to ensure that the bean bag could not fall forward. Furthermore, I adjusted the height and added my own 3D printed spacers to ensure that the claw could be attached to the gears firmly. Then, I decided to eliminate the fourbar and try a simpler approach by moving the claw mechanism to the bottom of the robot. This was mainly motivated by the fact that I was pivoting to a ramp, so I could have an intake that was fixed relative to the ground instead of a fourbar. However, before doing so, I experimented with my passive claw design as outlined in the next section.

Preliminary CAD of powered claw in a fixed position

**Key Learning Takeaways:**

- *What did you learn about your scoring mechanism?*
- *What did you learn about the materials you were making your robot from?*
- *What did you learn about manipulation of the bean bag?*
- *According to your learnings, what was your evaluation of your intended system?*
- *In light of your learnings, did you change your approach in designing and building your robot?*

For my scoring mechanisms, I learnt that even though the servo itself has very fine controls, integrating it with other systems makes it less precise, so it is hard to ensure that it does not stall. As for the fourbar, I learned that although the structure of it was simple, powering and moving it was a large issue.

From this claw design, I learned a lot about the manipulation of the bean bag. The most important lesson was that the bean bag only needed to be supported from two sides to be stable, and did not require anything in the middle for support - the bean bag could hold its shape relatively reliably, as long as no large forces were applied to it. I also realized that it would be difficult to manipulate the bean bag solely with sideways force, since the bean bag could quickly become impossible to deform further as it gets compressed. These results made me focus more on supporting the bean bag from two sides consistently, and not worry too much about squeezing the bean bag in place.

I believe this system worked, and would work even more reliably if I implemented the changes as mentioned above. I believe this robot would have worked fine as a final design if integrated with a ramp to drive onto the game board. However, in light of my learnings, I discovered that the underlying idea of it could be optimised further, as detailed in the next section.

## 2.3 Demonstration 3: Revision to scoring or acquisition system (8%)

**Integration Challenges:**

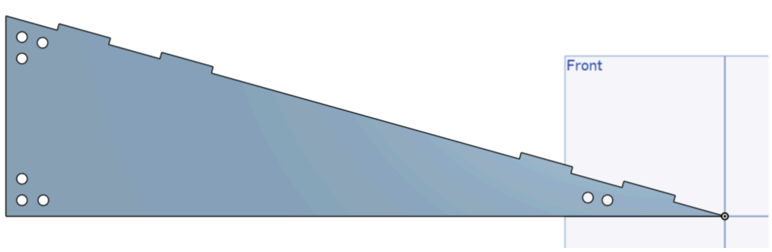
- *Which system did you demonstrate that you made a revision of?*
- *What failures from prior milestones lead you to select this system as the priority for revision over the other subsystems?*
- *What things did you change from the first version to the second? Why?*

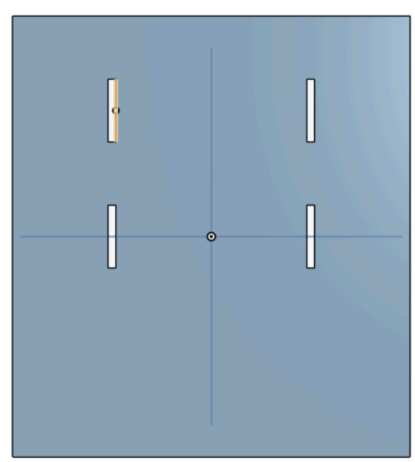
Revisions to getting onto the board:

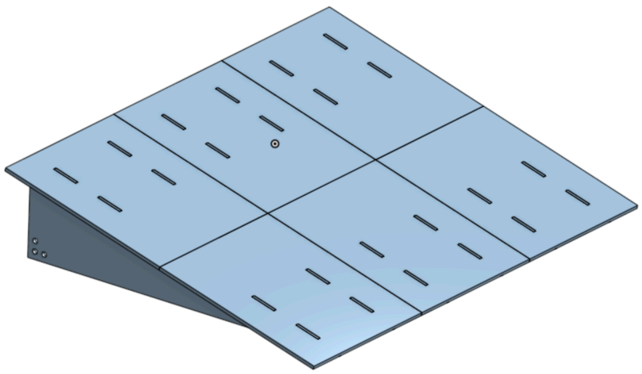
As mentioned before, my whegs meant that my robot was highly unstable: it turned very inconsistently and struggled to get onto the board smoothly as it lacked torque and grip. Furthermore, the change in the robot's height made it much harder for me to control my intake system and score points.

This revision was my first priority because the claw I demonstrated could pick up the bean bag consistently. Furthermore, getting onto the board has always been my main goal, which is also why it was my first milestone. Changes to my method of getting onto the game board would also affect my other subsystems, which was why I needed to revise my whegs first.

Therefore, I made the decision to turn to my backup plan of using a ramp to get onto the game board. My ramp would be made up of two components: triangles for the angle and base of the ramp, and rectangles to sit on top of the triangles to provide a smooth surface for the robot to drive over. I would then use 3” stand offs to connect each triangle for stability.



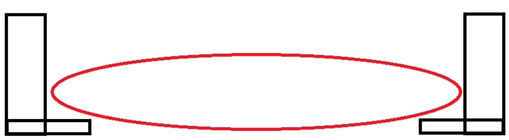





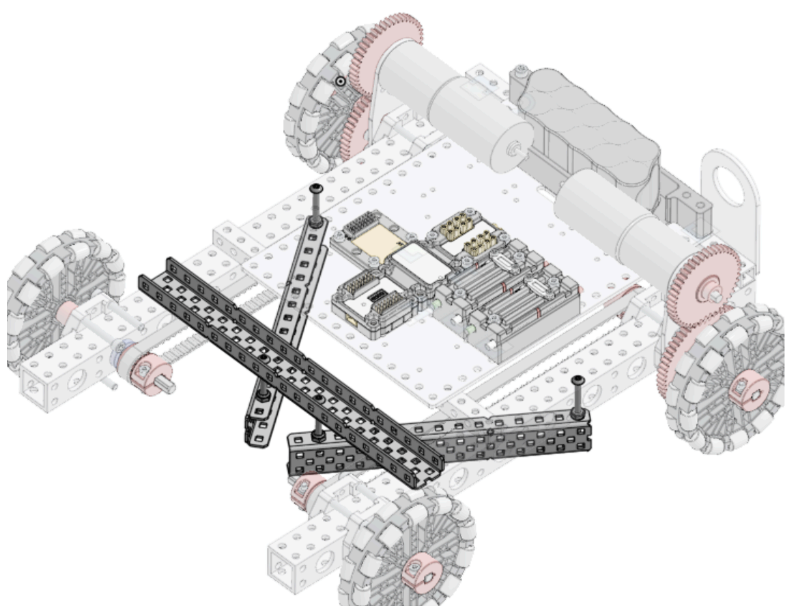
To construct the ramp, I decided to use ⅛" tempered hardboard that I purchased from Home Depot to build it since it could be laser cut and assembled together quickly. The main motivation for this was my experience with this material in the class Fundamentals of Mechanical Engineering (24-101), where we laser cut this exact material to build a hand powered flash light. This material was cheap, lightweight, rigid, and had both a smooth and rough side, which I planned on using to my advantage (for extra grip).

Revisions to intaking:

With my ramp made, I moved onto making a new intake system. This idea was not in my project proposal, but I had the idea to make a "plow" to move the bean bag. This idea largely stemmed from my claw, where I could support the bean bag securely by just contacting the other two edges, and did not need anything underneath the middle of the bean bag.



This gave me the idea to use two fixed c-channels to act as a funnel and a passive claw to manipulate the bean bag. The two c-channels would be slightly raised from the ground, forming a V-shape in the middle of the robot, giving me enough clearance to climb the ramp. Gravity would then push the bean bag into the scoring hole due to the lack of support at the front.



**Testing & Results:**

- *How did you test your system? Explain your setup and methodology.*
- *What worked well?*
- *Did you identify any failure modes and/or parts/components that underperformed? How?*
- *Provide video/photo evidence for your test and system.*
- *What was your plan to make your system perform better in the next demonstrations?*
- *Explain the problems and potential improvements in detail, e.g. insufficient power, inconsistency, gear failure, shaft interface failure, jamming,structural additions & modifications, control logic refinements,etc.*

To test the ramp, I drove the robot up and down it, making sure the ramp didn't move in the process. Furthermore, I also practiced using the robot to get the ramp back into place if I accidentally pushed it away from falling off the game ramp.

To test the claw, I simply drove into bean bags and made sure I could turn without losing control of it. Of course, since it was passive, I could not drive backward and still have the bean bag in my possession. The only slight inconvenience stemming from this was the fact that when intaking bean bags from the intake zone, I would need to drive behind the bean bag and then push it forward without going out of bounds. Since the intake zone was 3 feet long, this proved to not be a huge issue.

One major problem with the intake was jamming. Because my claw was fixed and near the ground, the distance between my claw and my wheels created small nooks that allowed bean bags to get stuck. If the bean bag was misaligned, it could easily get sucked into my gears/wheels when I drove over it, causing my robot to lift up and lose contact with the floor. The only way to get unstuck was to drive around randomly to try to shake it loose because my robot only had passive systems. In the end, I decided that this was not actually a major problem because we could only possess one bean bag at a time. Therefore, there would be no reason for me to have multiple bean bags on the field at once, and so I could focus on intaking one bean bag at a time.

The problems discussed here applied to each isolated system. Additional challenges arose when I combined these systems in the next section.

**Key Learning Takeaways:**

- What did you learn about the importance of iterating on a design?
- What ideas do you have that would help you speed up your design iterations?
- What was needed to improve on your robot?

Since both of these systems were completely new, I have moved these takeaways to the next section where I improve on these.

## 2.4 Demonstration 4: Intake/Scoring Integration (8%)

**Integration Challenges:**

- *How did you physically integrate intake and scoring?*
- *What were the new problems that emerged when combining systems?*
- *Did you have any interference issues, timing issues, power issues?*

The ramp and passive claw systems combined without any additional infrastructure, but physically integrating them caused new problems to emerge and required extensive fine-tuning.

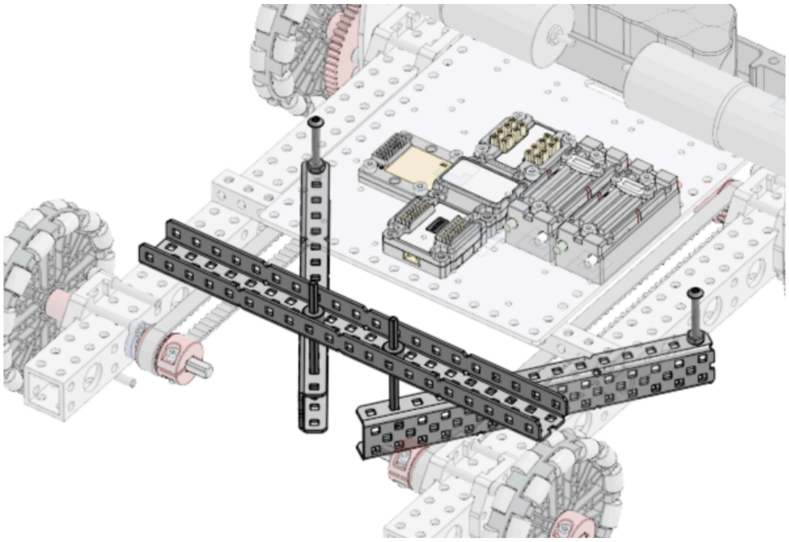
Since my ramp had individual panels that were unsupported where they met, they could shift and pop up a little as I drove on top, especially with the bean bag. This meant, as the bean bag got pushed onto the ramp, it could easily dislodge a panel, causing everything to get stuck. However, this was an easy fix as I could just replace the small panels with one large panel.

The most apparent problem for the claw was if the c-channels of the claw were too low, they would hit the edge of my ramp as I tried to drive off the game ramp. My solution was to change the spacing of the nyloc nuts that kept the c-channels in place. I continued to test and tune the spacings of the c-channels throughout the process of modifying the claw.

This led on to my biggest problem where the bean bag kept getting stuck underneath the c-channels when the robot went up the ramp. Since my c-channels were static, it was easier for the bean bag to get wedged underneath the c-channel, lifting my entire robot and losing contact with my ramp. As a result, I was unable to drive up the ramp anymore, and had to drive back down and try to catch the bean bag at a better angle. This proved to be an enormous problem that required many small insights to solve.

My natural instinct was that the space between the c-channels themselves (at the base of the V shape) was too large, allowing the bean bag to get wedged in very tightly. However, simply moving the c-channels closer together was insufficient as the bean bag could easily deform and wedge itself in the tiniest of gaps.

After observing the bean bag and robot as it climbed my ramp numerous times, I realised that when the system worked smoothly, the c-channels remained under the bean bag at all times. This was a result of the bean bag being in a specific orientation as it sat in the claw, something that I could not replicate consistently. This is where I had the idea to try to give the claw some flexibility. By replacing the back two screws that held the claw down with axles, the claw could now wiggle around. This proved to be the single greatest innovation of my design process as the tolerance gave the robot a much higher chance of hooking underneath the bean bag, especially when climbing my ramp.



**Testing & Results:**

- *How did your robot perform in the full cycle testing (pick up → score)?*
- *What worked well?*
- *Did you identify any failure modes and/or parts/components that underperformed? How?*
- *Provide video/photo evidence for your test and system.*
- *What was your plan to make your system perform better in the next demonstrations?*
- *Explain the problems and potential improvements in detail, e.g. insufficient power, inconsistency, gear failure, shaft interface failure, jamming,structural additions & modifications, control logic refinements,etc.*

With these changes, it worked very well, as seen in this demo: Section 2.4 Demo.mp4 .

At this point, I realised that the ramp I made was illegal because I laser cut the components. I manufactured a new ramp with the same dimensions except I did not use a laser cutter, and instead used a vertical bandsaw and some help from Tim in the Woodshop. I had to make some changes to the foundation of the design because I could not precisely cut holes in my large panel to slot in my triangles. Instead, I alternated wooden blocks with a width of 3 inches to act as the standoffs. More importantly, they provided a surface to nail the large panel on, allowing me to secure the panel onto the triangles. Note that the standoffs in the image are only there to secure the triangles as I glue and nail them together.

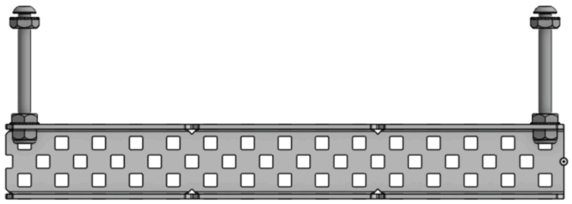
Since I was happy with the design, this was the design I used for the Final Demonstration. However, during the demonstration, I accidentally drove off the upper edge of the game ramp. I knew that the robot would get close to the edge to score the bean bag, but I did not realise it was such a fine line. I needed to concentrate on the distance of the robot from the edge at all times, making it very tiresome when doing the demonstration. Therefore, I planned on moving the passive claw forward to increase the distance between the back edge of the game ramp and the robot when scoring.

However, this proved to be another large obstacle. By moving the passive claw just two inches forward, the entire dynamics of the system changed. My hypothesis for this was that when the robot drove up my ramp, the bean bag and claw experienced a sharper turn because it was closer to the front. This made the entire system less reliable as it was more difficult for the passive claw to adjust to the incline.

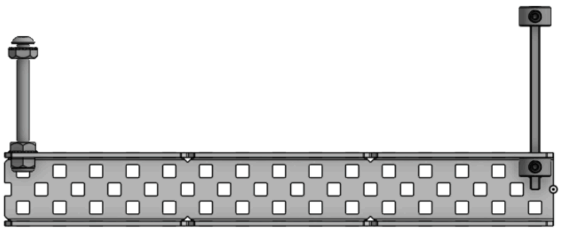
The solution I found in the end was a result of many experiments and iterations. First, I tried swapping the front two screws that held the claw with axles as well. I reasoned that having a sharper change in angle meant I needed a little more freedom for the claws to adapt to this change and maintain control over the bean bag. Then, with my passive claw being fully supported by axles, a new problem arose. With the extra freedom, the claw could be pushed very far back, allowing the bean bag to once again slip underneath the claw and get stuck. I experimented with the height of the claw, and changed shaft collar configurations to see how the height of the c-channels affected things. This did not work, so I tried to restrict the motion of the claw by adding additional shaft collars on top of the claw. This restricted how much it could tilt upwards without diminishing its horizontal freedom. This worked a lot better than before, and I again adjusted the shaft collar heights to achieve a good balance between the degree of tilt and height off the ground.

This culminated in a passive claw that was further forward, and only got stuck very few times. This I believed to be manageable because by simply driving back a few inches and then back forward again, the passive claw readjusted and easily lifted the bean bag up my ramp again, as seen in my second attempt during the Final Demonstration.

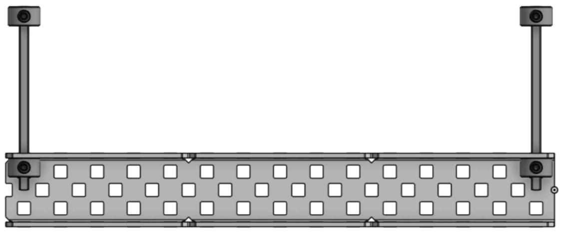
Below is the evolution of the supports for my claw from the beginning to end (left is front of claw):

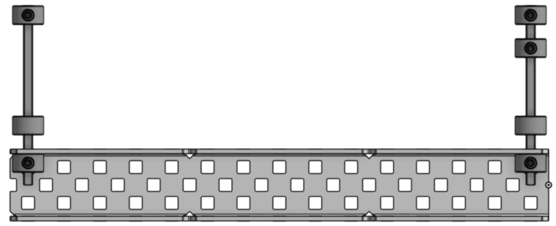


First Static Claw



Changed one pair of screws with axles





All supports are now axles

Added more shaft collars to restrict vertical freedom

**Key Learning Takeaways:**

- *What did you learn about the importance of iterating on a design?*
- *What ideas do you have that would help you speed up your design iterations?*
- *What was needed to improve on your robot?*
  *Explain what you learned about system integration.*
  - *What were surprises and challenges?*
  - *How would you plan differently in the future?*

In this stage, the importance of iterating on a design was strongly emphasized by my experimentation with the passive claw. Without iterating over designs, I would have had a much more unreliable system that may have achieved 20 points in five minutes, but in an unsatisfactory manner. Furthermore, it was important that I knew my first design would not be my final one, and that iterating on it should be expected. Lastly, making small iterations was also key, whether it was for my whegs to make sure that they were indeed unreliable, or for my passive claw where I had to carefully tune the configurations of my fasteners.

The testing process also taught me the importance of performing calculations to speed up design iterations. This was something that I neglected when making my fourbar. Doing preliminary calculations would have drastically reduced the number of iterations I had when making it.

To improve my design, I had to fully understand the physics of what was happening when my robot interacted with the bean bag. This enabled me to realise the need for more/less freedom when applicable, making my robot far more efficient and reliable than before.

During my system integration, I did not expect to encounter so many challenges. I underestimated how much the systems depended on each other because of how little they affected each other when building them. For something like a fourbar and a claw, it’s obvious that they heavily depend on each other, whether it's when you're building them or when using them. However, with my ramp and claw, there were no interactions between them when building them as they were separate entities. This made it harder for me to anticipate the challenges that arose when they were integrated together.

To mitigate these issues, it would be very beneficial if I could simulate the geometries of these mechanisms in 2D to see how they interact with each other. This could be possible in OnShape, but having a software for that purpose would make the physics very clear. Otherwise, making sure everything fits together in CAD would be a more general way to ensure that different systems can be successfully integrated.

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# Section 3: Critical Analysis & Reflection (30%)

## 3.1 Proposal vs. Final Design (10%)

**Original Proposal Review:**

- *Briefly summarize your initial design concept from the design proposal.*
    - *What were your key planned mechanisms and strategy?*

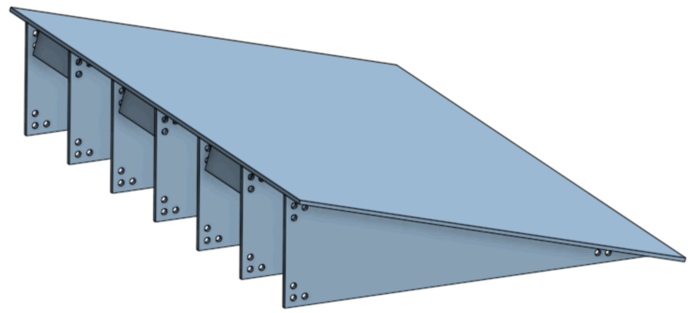
My initial design was to use whegs to climb the game ramp, then use rollers to intake and outtake bean bags. The whegs would be a special type of wheel that allowed me to hook onto the edge of the game ramp, lifting the robot up. Then, the rollers could intake and store the bean bag as the robot climbed onto the game ramp, allowing the robot to then outtake the bean bags into the cornhole board hole.

**What Changed and Why:**

- *For each major change from proposal to final design:*
    - *What did you originally plan?*
    - *What did you actually build?*
    - *Why did you make this change? (testing revealed, manufacturing constraints, time constraints, better ideas emerged, etc.)*
    - *Include CAD comparisons: proposal concept vs. final design*

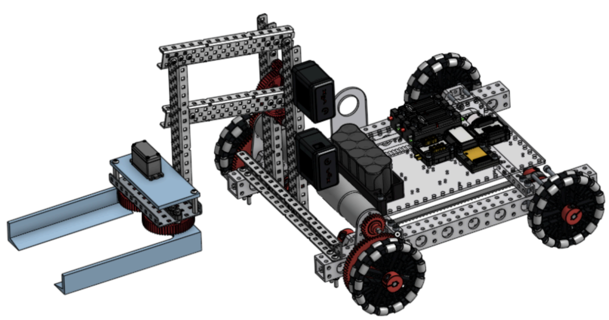
Wheg to Ramp:

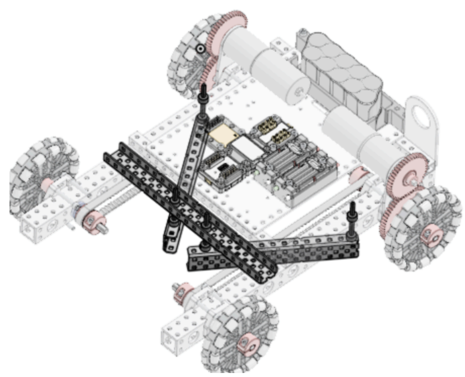
My method of getting onto the game ramp changed from whegs to a ramp. This change mostly stemmed from many unsuccessful iterations of the wheg, where its flaws far outweighed its potential upsides. The whegs made driving incredibly difficult as turning was inconsistent and the gait of the robot was very oscillatory. Furthermore, issues with grip and weight meant climbing the game board was mostly up to luck. The culmination of these issues resulted in my pivoting to my backup solution of a ramp.



Rollers to Claw:

As for my roller idea, I had to immediately switch to a claw and four bar design because of the vertical oscillation caused by my whegs. After eliminating the whegs, I designed a fourbar and powered claw, which eventually evolved into a passive claw that worked very well with my ramp. This evolution mainly came from experimentation with the shape of my powered claw and new ideas. I could have refined the design of the fourbar and claw instead, but it most likely would not have worked as well as the passive claw.





**Design Decision Analysis:**

- *Which changes were due to technical failures in testing?*
- *Which changes were due to strategic improvements?*
- *Which changes were due to constraints (time, materials, skill)?*
- *Which changes were due to re-reading and thinking of the competition rules?*

My change from whegs to a ramp was a result of technical failures in testing because they were not reliable and made everything else harder to integrate.

One change due to strategic improvements was going from a motorised claw to a passive claw. In my opinion, it was a lot easier to not have any extra actuators to worry about while driving, even though it meant my robot was a little less reliable. By having my concentration fully on driving, this made up for the extra effort needed to intake the bean bag with the passive claw.

The only change due to the task constraints was my need to remanufacture my ramp after I made it by using a laser cutter. A smaller change stemming from skill constraints was having to move my passive claw forward, so that I did not need to get as close to the edge of the board and risk driving off it to score the bean bag.

I had no changes that were due to re-reading the competition rules.

## 3.2 Final Design Evaluation (10%)

**What Worked Well:**

- *Which subsystems performed reliably?*
- *Which design decisions were most successful?*
- *What calculations/analysis were most useful?*

My ramp was the most reliable and successful subsystem. The geometry was precise which provided a smooth way for my robot to get on and off the game ramp, allowing me to go at higher speeds up and down the ramp. Furthermore, it did not need extra weights to prevent it from sliding around when in use, making it light and simple.

My most successful design decision was my insight into the mechanics of my passive claw. This was more a result of experimentation, but my idea of making the passive claw able to move around to accommodate the changing geometry of the bean bag proved to be very innovative.

I think my most useful calculation was the dimensions of my ramp. Even though it was relatively simple, having good geometries for my ramp allowed my passive claw idea to flourish. Without a well made ramp, I believe I would have struggled a lot more with experimenting with claw designs, as the geometries of my ramp could have profound effects on the effectiveness of my claw/intake mechanism.

**What Didn't Work:**

- *Which subsystems were problematic?*
- *What would you redesign given more time?*
- *What calculations or assumptions were wrong?*

Interestingly enough, I think my most problematic subsystem was the drive base itself. More specifically, the electronic components took up valuable space and affected my field of vision. The plastic plate that the electronics were bolted to took up space that could have been used by my passive claw. Furthermore, it considerably blocked my view of the claw and bean bag, so it was hard to tell whether or not the bean bag was scored.

Given more time, I would play around with the configuration of the electronics more, such as trying to mount them on the sides. This would naturally lead me to play around with the weight distribution of the robot and potentially enable me to move the claw further forward. Otherwise, there wasn't much else I could redesign.

I had a few wrong assumptions that made it much more difficult to make stable subsystems. The most critical one was when I assumed the whegs would not affect other subsystems. This meant my proposed intake design of using rollers was not feasible, so I had to immediately turn to my backup plan of using a claw.

Another mistake I made was the lack of preliminary calculations. When making the fourbar, I did not use sufficient calculations to determine how much torque I needed to lift the claw and bean bag, meaning I had to iteratively increase the gear ratio and add another motor, which cost me a lot of time.

Lastly, I wrongly assumed that I was allowed to use Techspark resources, like the laser cutter, which I should have gotten clarification on before deciding to lasercut my ramp.

**Performance vs. Goals:**

- *Did you meet your original success criteria?*
- *Where did your robot exceed expectations?*
- *Where did your robot fall short?*

I was able to exceed my original success criteria outlined in my proposal, which was to “attempt to score each of the 15 cornhole bags, of which I want to try to score 10/15 in the hole, and 5/15 either unscored or on the board”. Instead, I was able to score 15 cornhole bags in the hole in 4 minutes and 30 seconds.

My robot exceeded my expectations by being so simple, since I used no extra actuators other than the two included in the drive base. This made it extremely easy to control and manipulate the robot when driving.

My robot fell a little short in terms of how quickly I could drop the bean bag into the hole. In my first demonstration, I accidentally drove too far and fell off the game ramp, causing me to ruin my run. This meant I had to be more careful when at the top of the game ramp to make sure I didn’t fall off again, slowing me down considerably. However, this didn’t matter too much as I was still able to score almost all the bean bags in under five minutes in my second demonstration.

## 3.3 Lessons Learned & Recommendations (10%)

**Technical Lessons:**

*If you were to start over, what kind of advice would you give to yourself from the insights you have regarding:*

- *Your original plan?* Definitely do more analysis on how different subsystems affect each other, especially torque calculations.
- *The changes you made over your original plan?* Expect that your plans will change, and be receptive to change. Budgeting time for this can make it a lot less stressful when implementing changes.
- *Design?* Try to keep it simple. If something requires large changes over and over to make it work, it may not be worth it or be the best solution.
- *Manufacturing?* CAD before manufacturing and plan out how you would create each element, including what tool to use and how to orient the piece when machining it. This made it much quicker for me to manufacture and iterate designs, so more time could be spent testing and experimenting.
- *Mechanisms?* I think the most important advice is the same as design: keep them simple. The more complex it gets, the harder it'll be to fix and to manage.
- *Time management?* Starting early is best, especially to avoid last minute part manufacturing in the queue.

**Process Lessons:**

- *What do you think is the value in iterative testing or in trying to design a final product from the get go?* I think both have benefits and it depends more on the situation. For me, the ramp was more of a final product I made from the get go. This was mainly because the geometry was simple and very constrained, as the requirements were very precise and easily measureable. This meant there was less room for iteration, which saved time. However, iterative testing was also very useful. My passive claw was a result of iteratively testing different configurations, mainly because the geometries involved were not as constrained as the ramp, especially with the unpredictable nature of the bean bag orientation. Therefore, both methodologies are valuable.
- *What is the difference between building with or without CAD?* Similar to my points above, building with a CAD is much more useful when dealing with designs that have concrete constraints. When these constraints are relatively unknown, or can only be found out through experimentation, CAD is not as useful.
- *What did you learn about time management?* I believed I managed my time well, as I was on-time for every milestone and did not need to work late into the night. I did this by consistently scheduling small chunks of time to work on the robot throughout the week, making sure to dedicate more time at the beginning of the week.

- *What did you find to be the most frustrating thing about building a robot?* The most frustrating part of building a robot is the need to constantly disassemble and reassemble the subsystems. As I iterated through different strategies and design, I spent more time disassembling old parts compared to building new parts. In fact, my final passive claw that I designed could have very easily been built within just a few minutes as it was only composed of three C-channels and some axles and fasteners.
- *What did you find to be the most enjoyable and rewarding thing about building a robot?* The most enjoyable thing is obviously seeing it work in real life, especially the fact that it worked so well. It's amazing to think about the progression I made from the start of this project, and how much better my final system was compared to the beginning.

**Advice for Future Students:**

*Assume the final project will be revised so these are pieces of advice that you are giving **regarding designing and building robots in general**, and not specifically for the "payload pitch game".*

- *What should they focus on first?* Make sure to understand all of the rules and restrictions fully. This allowed me to come up with better ideas that made my robot design simpler and more efficient.
- *What are some mistakes they should be careful about/avoid?* I believe the most important mistake they should avoid is making incorrect assumptions. It's very important to double check your assumptions and try to get a second opinion on them in case you missed something important.
- *What resources should they take advantage of that you regret not doing or doing earlier?* Meetings with Tim. Discussing with Tim outside of the weekly meetings helped me to talk through ideas and either develop them or find better ones, especially in terms of manufacturing processes.
- *Would you recommend or warn against any mechanism types? Why?* I don't think people should avoid any specific mechanism types. However, in terms of general mechanisms, keeping it simple is always best. Having simple mechanisms makes it much easier to fix and improve, which greatly increases the amount of experimenting you can do.
- *How can they use the milestones or demonstrations to their advantage?* I think students can use milestones and demonstrations as a good source of motivation. Personally, I stressed a lot over milestones, which wasn't fun, but it provided a very good way for me to stay motivated and constantly improve.

# 8. References

You can refer to the following Data Sheets for material properties of PETG  and TPU for your design load calculations

FlySky  Controller Manual

Cite any additional resources, inspiration sources, or technical references used in the creation of your robot or this report.

Resources and references have been cited within the report.